

Large-Scale Convex Optimization

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Convexity

Def. f convex $\iff \text{epi}(f)$ convex $\iff f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y) \forall \theta \in [0, 1]$.

Characterizations (use these):

- Twice diff.: $\nabla^2 f(x) \succeq 0 \forall x$ (evals for $x^\top Q x$).
- 1st-order: $f(y) \geq f(x) + \nabla f(x)^\top (y - x) \forall x, y$.
- Monotone grad: $(\nabla f(x) - \nabla f(y))^\top (x - y) \geq 0$.
- $f^{**} = f$ (proper, closed epi).
- Restriction to any line is convex.

Preserving ops: nonnegative lin. comb., pointwise **max**, affine comp. $f(Ax + b)$, intersection of convex sets.

Does **NOT** imply convex (traps + counterex.):

- Unique/global or single local min ($f(x) = |x|$ on $[-1, 1]$, 1 outside).
- $\nabla^2 f$ bounded below (must be $\succeq 0$).
- $|\nabla f(x) - \nabla f(y)| \geq \mu |x - y|$ ($\mu > 0$).
- Pointwise **min** of convex: e.g. $f_1 = (x+1)^2$, $f_2 = (x-1)^2$.
- Product $f_1 f_2$: e.g. $x \cdot (-x) = -x^2$ (concave).
- $\exp(f(x))$ or $g(x) = 1/\ln x$ ($x > 1$) (Hessian neg.).
- $\sqrt{x}$ on $[0, 1]$: no finite Lip. const. ($\nabla f \rightarrow \infty$ at 0).

Examples:

- $\frac{1}{2}x^2 + p \ln x$ on $[1, \infty)$: convex $\iff p \leq 1$ ($\nabla^2 = 1 - p/x^2 \succeq 0$).
- $1/x_i$ convex on $\{x > 0\}$.
- Quadratic $x^\top Q x$ convex $\iff Q \succeq 0$.

Tactic (MC/TF): Test Hessian/1st-order or give counterex. Local min = global $\nRightarrow$ convex.

Conjugate $f^*(y) = \sup_x (y^\top x - f(x))$

f proper convex closed:

$$f^{**} = f. \quad y \in \partial f(x) \iff x \in \partial f^*(y).$$

4-Step Recipe:

1. Set $\nabla_x (y^\top x - f(x)) = 0 \implies \nabla f(x) = y$.
2. Solve for stationary x_y .
3. If $x_y \in \text{int}(\text{dom } f)$, $f^*(y) = y^\top x_y - f(x_y)$.
4. Restricted domain $[a, b]$: evaluate boundaries ($x_y < a \rightarrow x = a$, $x_y > b \rightarrow x = b$).

Quiz Examples:

- $f(x) = -x^2, x \in [-1, 1]$: $f^*(y) = 1 + |y|$ (V-shape plot, label axes).
- $f(x) = x^2 + x, x \in [-1, 1]$: stationary $x = \frac{1}{2}(y - 1)$,

$$f^*(y) = \begin{cases} -y & y \leq -1 \\ \frac{1}{4}(y - 1)^2 & -1 \leq y \leq 3 \\ y - 2 & y \geq 3 \end{cases}$$

- $f(x) = -\ln x (x > 0)$: $f^*(y) = -1 - \ln(-y) (y < 0), \infty$ else.
- $f(x) = e^x$: $f^*(y) = y \ln y - y (y > 0), \infty$ else.
- $f(x) = \frac{1}{2}x^\top Qx$: $f^*(y) = \frac{1}{2}y^\top Q^{-1}y$.

Tactic: Plot if asked (label axes). Stationary often quadratic, always check boundaries.

Subdifferential & Optimality

$$\partial f(\bar{x}) = \{v \mid f(x) \geq f(\bar{x}) + v^\top (x - \bar{x}) \forall x\}.$$

Optimality: $0 \in \partial f(x^*) \iff x^*$ global minimizer (unique if strict convex).

Example $f(x) = |x|$:

$$\partial f(x) = \begin{cases} \{-1\} & x < 0 \\ [-1, 1] & x = 0 \\ \{1\} & x > 0 \end{cases}$$

Minimizer at points where subdifferential plot contains 0 (vertical segment over $y = 0$).

Tactic (plot Q): Find point where subdiff. interval/segment includes 0.

Projections (PGD): $x_{k+1} = \text{prox}_{T,C}(x_k - T\nabla f(x_k))$.

- $\|x\|_\infty \leq 1$: clamp each coord to $[-1, 1]$.
- $\|x\|_2 \leq 1$: $z / \max(1, \|z\|_2)$.
- ℓ_1 -ball: soft-thresholding.

Duality, Lagrangian, KKT

Lagrange Duality

Consider $f^* = \inf_{x \in \mathbb{R}^n} f(x)$ s.t. $g(x) \leq 0, h(x) = 0$ (1)

Lagrangian $\mathcal{L}(x, \lambda, \nu) = f(x) + \lambda^T g(x) + \nu^T h(x)$

Dual Function $d(\lambda, \nu) = \inf_{x \in \mathbb{R}^n} \mathcal{L}(x, \lambda, \nu)$

Weak duality: $d^* \leq p^*$ always.

Strong duality + KKT (convex + Slater: $\exists \hat{x}, g_i(\hat{x}) < 0$):

$$\begin{aligned} \nabla_x \mathcal{L}(x^*, \lambda^*, \nu^*) &= 0, \\ g(x^*) &\leq 0, \quad h(x^*) = 0, \quad \lambda^* \geq 0, \\ \lambda^{*\top} g(x^*) &= 0 \text{ (compl. slackness)}. \end{aligned}$$

KKT Conditions (Karush-Kuhn-Tucker)

If Slater's condition holds and OP is convex $\rightarrow$
 $x^* \in \mathbb{R}^n$ is a minimizer of the primal and $(\lambda^* \geq 0, \nu^*) \in \mathbb{R}^{n_g} \times \mathbb{R}^{n_h}$ is a maximizer of the dual $\Leftrightarrow$ to the following statements:

KKT-1 (Stationary Lagrangian) $\nabla_x \mathcal{L}(x^*, \lambda^*, \nu^*) = 0$

KKT-2 (primal feasibility) $g(x^*) \leq 0, h(x^*) = 0$

KKT-3 (dual feasibility) $\lambda^* \geq 0, \nu^* \in \mathbb{R}^{n_h}$

KKT-4 (complementary slackness) $\lambda^{*\top} g(x^*) = 0$

$\nu^{*\top} h(x^*) = 0$

In addition we have: $\sup_{\lambda \geq 0, \nu \in \mathbb{R}^{n_h}} q(\lambda, \nu) = \inf_{x \in C} f(x)$

How to solve any KKT problem (exam tactic):

1. Write problem as $\min f(x)$ s.t. $h(x) \leq 0, g(x) = 0$ (ensure ≤ 0 format).

2. Write Lagrangian: $\mathcal{L}(x, \lambda, \nu) = f(x) + \lambda h(x) + \nu g(x)$.
3. List the 4 KKT conditions (as above).
4. Solve by branching on complementary slackness:
 - Case A: Assume $\lambda = 0$ (constraint inactive), solve $\nabla_x \mathcal{L} = 0$ for x , check if $h(x) \leq 0$.
 - Case B: Assume $\lambda > 0 \implies h(x) = 0$ (constraint active), use this equality to solve for x, λ , check $\lambda \geq 0$.

Tactic: Write full $\mathcal{L}$, set $\nabla_x \mathcal{L} = 0$, substitute x^* into $d(\lambda)$, maximize (often quadratic). Check Slater. Compl. slackness key.

Example $\min \frac{1}{2} \|a\|^2$ s.t. $y_i(a^\top x_i + b) \geq 1$; dual $\max_{\lambda \geq 0} d(\lambda)$. Solves to $\lambda_A^* = \lambda_B^* = 1/4$.

Example $f = 1/x_1 + 4/x_2, x_1 + x_2 \leq 3, x_{1,2} \geq 1$: write full $\mathcal{L}$, solve stationarity+slack.

ADMM Derivation (mechanical steps)

Step 1: Reformulate problem to have separated variables:

$$\min_{x,z} f(x) + g(z) \quad \text{s.t. } Ax - z = c.$$

Step 2: Write Augmented Lagrangian:

$$\mathcal{L}_\rho(x, z, \lambda) = f(x) + g(z) + \lambda^\top (Ax - z - c) + \frac{\rho}{2} \|Ax - z - c\|_2^2.$$

Step 3: Update x : Set $\nabla_x \mathcal{L}_\rho = 0$. This usually results in a linear system:

$$(A^\top A + \text{something})x = \dots$$

Step 4: Update z : If $g(z)$ is an L_1 norm (e.g., $|z|_1$), the solution is exactly the proximal operator/soft-thresholding:

$$z_{k+1} = \text{prox}_{1/\rho}(Ax_{k+1} - c + \lambda_k/\rho).$$

Step 5: Update λ :

$$\lambda_{k+1} = \lambda_k + \rho(Ax_{k+1} - z_{k+1} - c).$$

Tactic: Memorize the 5 steps; the x -update reduces to solving a linear system; the z -update is often a closed-form proximal operator.

Reformulating max and absolute values (Epigraph)

When minimizing $\max(f(x), g(x))$ or $|f(x)|$:

- Introduce an auxiliary variable t .
- New objective: $\min t$.
- For max: constraints are $f(x) \leq t$ and $g(x)$